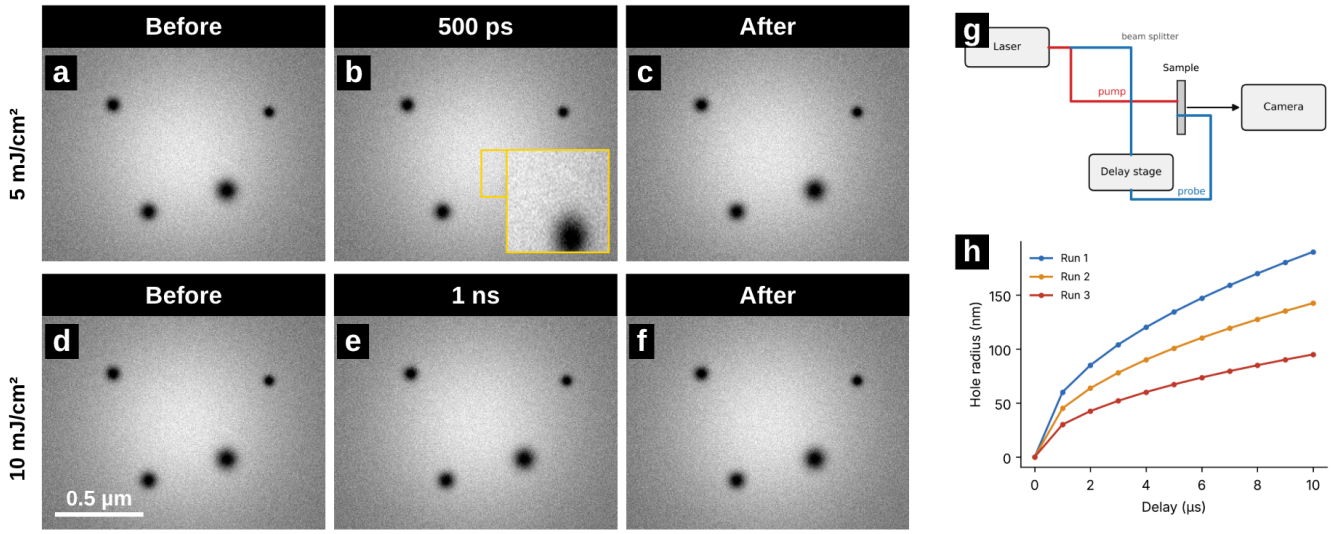
TIMEFRAME STUDIO

User Guide

Publication-ready figures from your Before, time-resolved and After microscopy images — arranged on a free canvas, lined up like in PowerPoint, and exported losslessly. Your pixels are never changed.



A figure made in TimeFrame Studio: two runs in a 2 × 3 grid with titles read from the file names, fluence labels, panel letters, a calibrated scale bar, a zoom inset, and a schematic and a plot lined up beside the grid. Exported at 300 dpi for this page. (Sample images.)

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01 What TimeFrame Studio does

It replaces the PowerPoint → PDF → Preview "Export as TIFF" routine with one tool built for microscopy figures. You bring the photos; it arranges, labels, lines up and exports them.

- **Your photos are never altered.** Brightness, contrast, crop and background removal are display settings only. The original files are kept, byte for byte, inside your project file.
- **A free canvas, like a slide.** Put photos, grids, text and shapes anywhere. A **frame** that you size yourself marks what gets exported — it never grows on its own.
- **BrightSlide-style arranging.** Click the reference photo first, Shift-click the others, then align them, make them the same size, space them or put them in a grid.
- **Time-series grids, optional.** MAKE GRID orders photos Before → delays → After from their file names and titles them ("500 ps", "1.5 µs"…) for you.
- **Formulas in any text**: type `\Delta t = 1.5 \mu s` or `J/cm^2` the LaTeX way; it is drawn in your figure's own font.
- **Gatan DM3 / DM4** files open directly, with their pixel size, next to TIFF, PNG and JPEG.
- **Groups, turning and flipping**, a gap you type between two things, and matching only the width or height of a reference.
- **Lossless export** to TIFF, PNG, PDF or SVG at the size and dpi you choose — a **native 1:1** mode that copies camera pixels exactly — and an **editable PowerPoint** slide.
- **Batch**: one figure per experiment folder, all in the same style, exported in one ZIP.

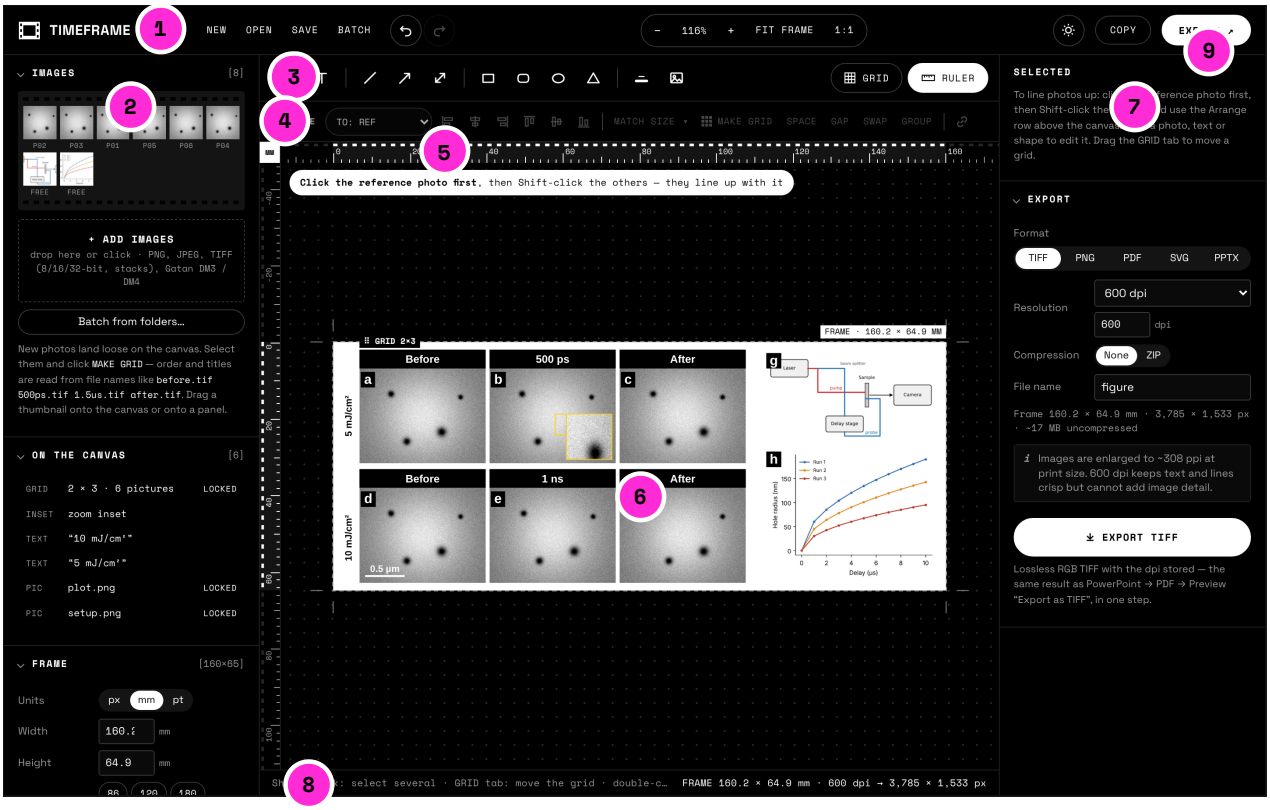
RUNS ANYWHERE

TimeFrame Studio is a single HTML file. Double-click it to open it in Chrome, Edge, Firefox or Safari — nothing to install, and nothing is uploaded: your images stay on your computer.

02

Quick start and the screen

1. **Set the frame.** In **FRAME** (left) pick a journal width (86, 120 or 180 mm) or type the width and height you want.
2. **Drop your photos** on the canvas (or click **+ ADD IMAGES**). They land loose and selected.
3. **Make a grid** for a time series: click **MAKE GRID** in the Arrange row. Order and titles come from the file names; fix anything shown in pink on the right.
4. **Line other things up**: click the reference first, Shift-click the others, then use the align buttons, **MATCH SIZE** or **SPACE**.
5. **Add text, arrows, letters and a scale bar** from the toolbar and the left panel.
6. **Check and export**: read the checks above **EXPORT TIFF** (right), then export. Click **SAVE** to keep an editable project.



The screen in the dark theme (the sun/moon button switches to light). Numbers refer to the list below.

1. **File and history** — NEW, OPEN, SAVE, BATCH, undo / redo.
2. **Left panel** — images, what is on the canvas, frame, alignment grid, time-series grid, titles, letters, scale bars, lab style.
3. **Toolbar** — select, text, lines, arrows, shapes, scale bar, insert image; GRID and RULER switches.
4. **Arrange row** — align, match size, make grid, space, swap, lock. Always visible; greyed out until something is selected.
5. **Rulers** — zero sits on the frame's top-left corner. Drag from a ruler to add a guide.
6. **Canvas** — the white **frame** is what gets exported; the dotted area around it is free room.
7. **Right panel** — settings for what you selected, and Export.
8. **Status bar** — tips, and the frame size in mm and in exported pixels.
9. **Zoom, theme, COPY and EXPORT.**

03

Bringing in photos

Drop image files anywhere on the canvas, or click **+ ADD IMAGES**. PNG, JPEG and TIFF are read — including 16- and 32-bit TIFF, LZW / ZIP / PackBits compression and multi-page stacks (each page becomes its own image) — and **Gatan DM3 / DM4** files (below).

- **New photos land loose**, in a tidy row, and stay selected — ready for MAKE GRID or for aligning. Nothing pops up.
- **Dropped onto a grid panel**, a photo goes into that panel (several photos fill the panels from there).
- **The library** shows every photo you have brought in. Click a thumbnail to select that photo; click an unused one to place it in the middle of the view; or drag a thumbnail onto the canvas or onto a panel.



The library. **P01** = in panel 1 of the grid, **FREE** = loose on the canvas, **—** = not used yet.

What is read from each file

FROM	WHAT	USED FOR
File name	`before`, `pre`, `ref`, `initial`, `unpumped` → **Before** `after`, `post`, `final`, `recovered` → **After** a number with fs, ps, ns, µs/us, ms or s → a **delay** (“500 ps”, “1.5 µs”) `TR` alone → time-resolved, delay unknown	Order (Before → shortest delay → longest → After) and titles
File name	What is left once those words are removed (e.g. `Run01`, `Run02`)	One run per row in MAKE GRID
TIFF metadata	Pixel size from ImageJ, OME and FEI / Thermo files	Calibrated scale bars
DM3 / DM4	The image itself (the preview thumbnail is skipped), 8- to 64-bit integer or float, either byte order; its pixel size in pm, Å, nm, µm or mm	Calibrated scale bars

GATAN DM3 / DM4 **NEW**

Drop them like any image. A **stack** (an in-situ series) gives one picture per frame, up to 200, named `file.dm4 [1]`, `[2]` … so you can pick frames for a time series. Things that are not plain images are refused with a short note: 1-D spectra, spectrum images (EELS / EDS / CL), FFT / complex data and 4D-STEM — export the image you need from DigitalMicrograph first. Diffraction patterns open, but their reciprocal-space calibration (1/nm) is not used for scale bars.

TIP

Name files like `Run01_before.tif`, `Run01_TR_500ps.tif`, `Run01_after.tif` and the grid builds itself: one row per run, columns in time order, titles filled in.

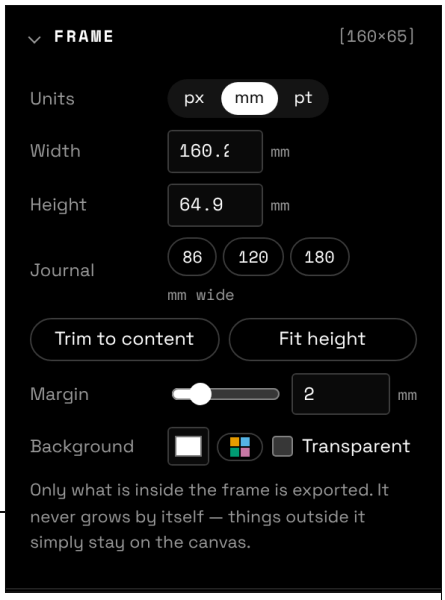
04

The canvas and the frame

Think of the canvas as a desk and the frame as the sheet of paper. Only what is inside the frame is exported. Things can wait outside it — they stay on the canvas and are saved with the project.

Frame size

- **Width and Height** — type them in mm, px or pt (the Units switch; px means pixels in the exported file at the current dpi).
- **Journal** — one click for 86, 120 or 180 mm wide (1, 1.5 and 2 columns at most journals).
- **Trim to content** — the frame shrinks or grows to hug everything on the canvas, plus the margin. Its size is rounded to whole exported pixels.
- **Fit height** — keeps your width and fits only the height.
- **Margin** — the empty border used by Trim and by MAKE GRID's "fit frame".
- **Background** — any colour, or Transparent (kept in PNG and TIFF).



FRAME in the left panel.

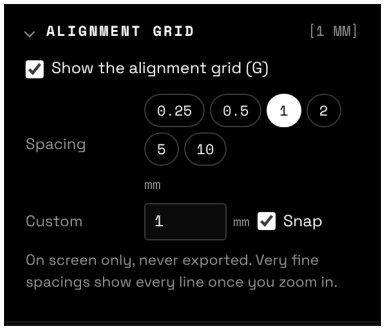
> THE FRAME NEVER GROWS ON ITS OWN
>
> If something crosses the frame edge, the checks above the Export button say so ("1 item crosses the frame edge and will be cut"). Move it in, change the frame size, or use Trim to content.

Moving around

- **Zoom**: ⌘/Ctrl + scroll zooms around the pointer; − and + in the top bar; **FIT FRAME** shows the whole frame; **1:1** shows the printed size.
- **Scroll** with the mouse wheel or trackpad. The canvas grows around anything you place outside the frame.

Rulers, guides and the alignment grid

- **RULER** (⇧R) shows rulers whose zero is the frame's top-left corner. The corner button switches mm / px / pt.
- **Guides**: drag from a ruler onto the canvas. Drag a guide back off the canvas to remove it.
- **GRID** (G) shows the alignment grid. Choose 0.25, 0.5, 1, 2, 5 or 10 mm, or type your own spacing. Tick **Snap** to make things snap to it. When lines are too close to see, only every 5th (or 25th) is drawn until you zoom in.
- Rulers, guides and the alignment grid are **on screen only** — never exported.



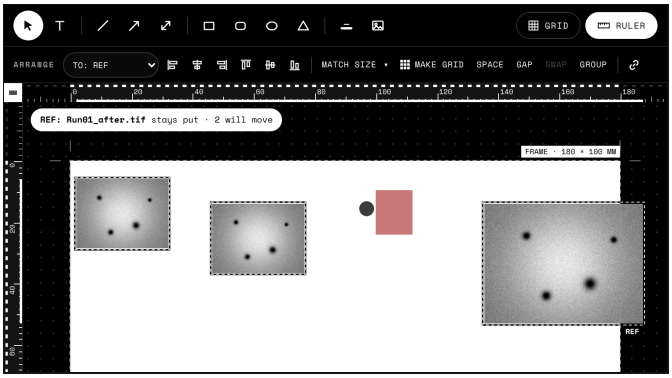
ALIGNMENT GRID: spacings from 0.25 mm, or any value.

05

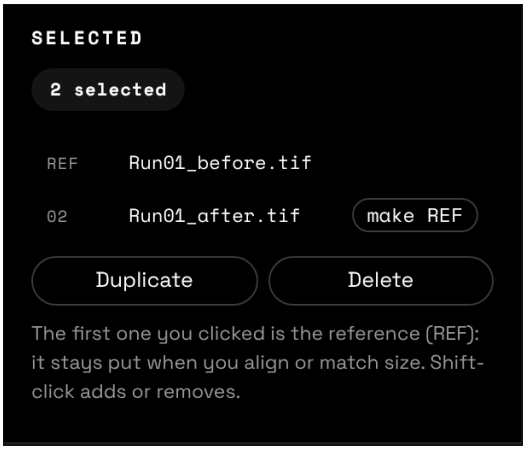
Selecting — and the reference

Arranging works the way BrightSlide does in PowerPoint: the first thing you click is the **reference**. It stays put; everything you add after it moves to match it.

TO	DO THIS
Select one thing	Click it.
Add or remove things	`Shift`-click (or `⌘`/`Ctrl`-click). The first one stays the reference.
Select by area	Drag a box on empty canvas. `Shift`-drag adds to what is already selected. The grid joins when the box covers all of it.
Select everything	`⌘A` / `Ctrl A`
Pick one out of several	Click it without dragging: only it stays selected.
Use a grid panel as the reference	Click the panel first, then Shift-click the photos to line up with it — or Shift-click the panel.
Change the reference	Click **`make REF`** next to another item in the right panel.
Select the whole grid	Click its ⠿ **`GRID`** tab, or GRID in “On the canvas”.



The reference carries a black **REF** label; the hint at the top of the canvas names it and says how many things will move.



The right panel lists the selection, reference first, with **make REF** buttons.

> “ON THE CANVAS” LIST
> The left panel lists the grid and every object, top to bottom. Click an entry to select it, Shift-click to add it. Items with a locked alignment are marked **LOCKED**.

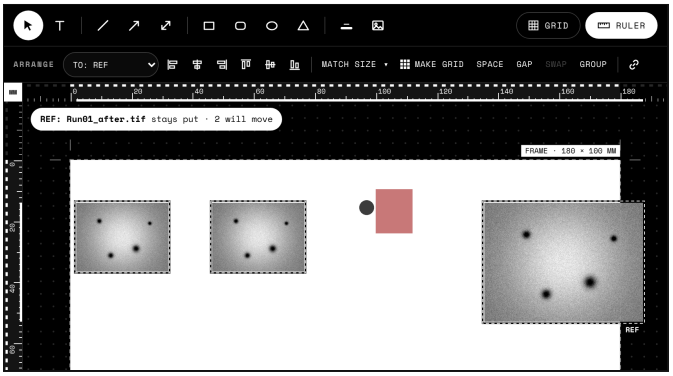
06

The Arrange row

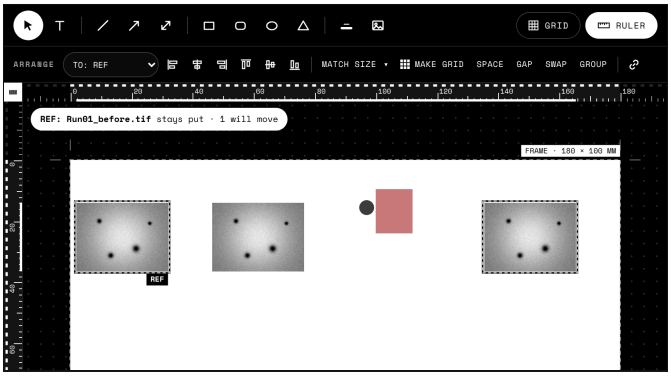


The Arrange row sits under the toolbar and is always there. Buttons switch on as soon as they can do something.

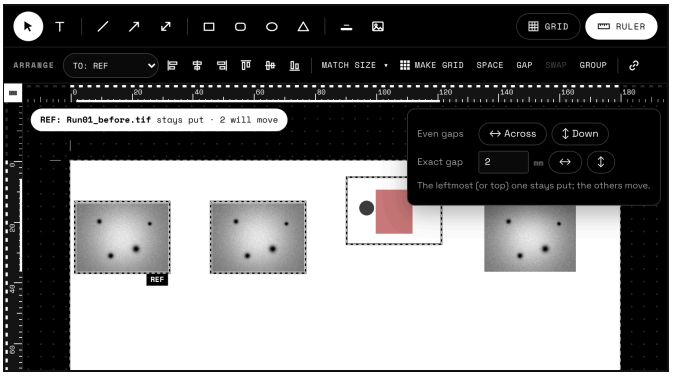
BUTTON	WHAT IT DOES
TO: REF ▾	What to line up with: **REF** (the one you clicked first), the whole **SELECTION** (its outer edge), or the **FRAME** (inside its margin). With one thing selected it lines up with the panel it sits on, or the frame.
Six align icons	Left edges · centres · right edges · top edges · middles · bottom edges.
MATCH SIZE ▾	Makes the others exactly as wide and as tall as the reference. Photos are **cropped to fill** — never stretched. Shapes are resized; a grid makes its panels that size. The ▾ beside it offers **Width only** or **Height only** NEW: the button then reads MATCH WIDTH / MATCH HEIGHT.
MAKE GRID	Puts the selected loose photos into rows and columns (chapter 07).
SPACE	**Even gaps** across or down (three or more things), or an **exact gap** you type (two or more). The leftmost or top one stays put.
GAP NEW	Type the distance between the **facing sides** of the reference and the other one — e.g. the grid's bottom and a picture's top. See below.
SWAP	Two things change places. Two panels swap images; a loose photo and a panel exchange images.
GROUP NEW	Makes the selected things one piece (⌘G); with a group selected it reads **UNGROUP** (⇧⌘G). Chapter 11.
🔗 Lock	Keeps an alignment. Click it **before** an align button, or **right after** aligning. Locked things follow their reference when it moves, and slide only along the locked edge.



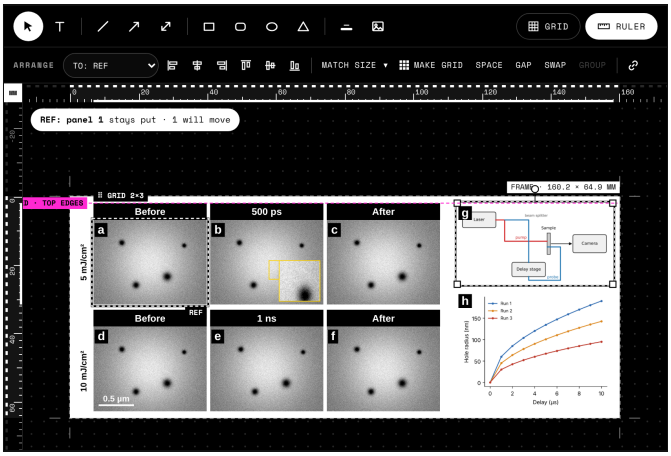
Align top edges with the big photo as REF: it stays, the two others move up to its top.



Match size: the second photo now has exactly the reference's width and height.



Space: equal gaps, or an exact gap in mm, px or pt.

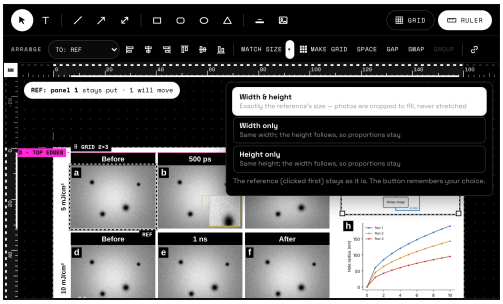


A **lock**: the schematic's top edge stays on the grid's top edge (pink line). Unlock it in the right panel.

Match width or height only NEW

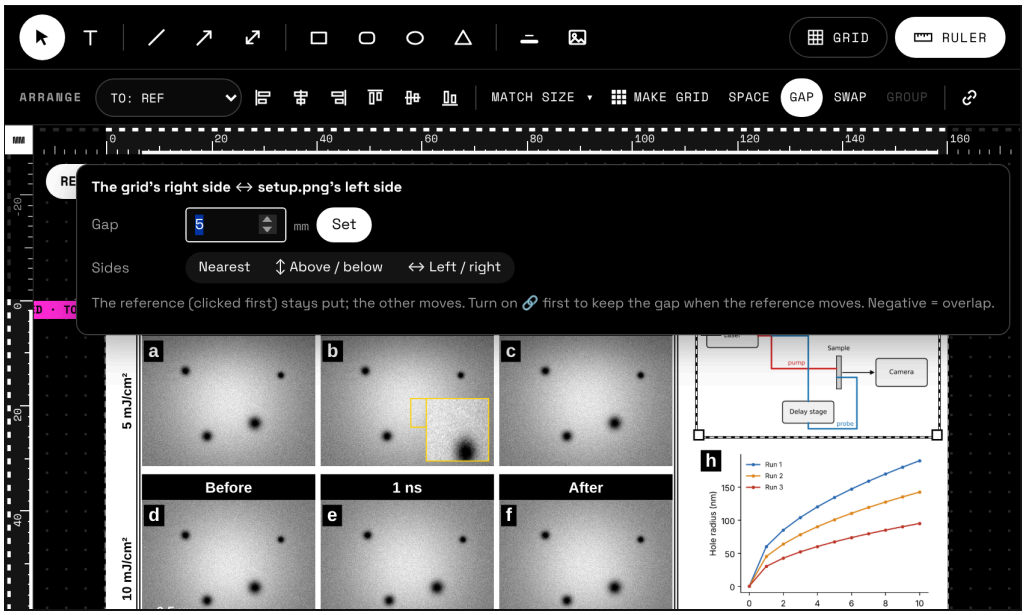
Click the reference first, Shift-click the others, then open the ▾ next to MATCH SIZE:

- **Width only** — each one gets the reference's width; its height follows, so its **proportions stay** and nothing is cropped or stretched.
- **Height only** — the same for the height, e.g. a row of pictures of different shapes, all as tall as the first.
- Works on photos, pictures, shapes, text (its size changes), groups and the grid (the whole grid zooms, text and scale bars included). Each keeps its top-left corner.
- The button remembers your choice: it then reads **`MATCH WIDTH`** or **`MATCH HEIGHT`** and works with one click. Pick **Width & height** to go back.



The ▾ next to MATCH SIZE: width & height, width only, or height only.

A gap you type (GAP) NEW



The grid (clicked first) and a schematic beside it: GAP shows the distance between the grid's right side and the schematic's left side, ready to change.

1. Click the reference (it stays put), then Shift-click the one to move. It can be a photo, a picture, text, a shape, a group or the grid.
2. Click **GAP**. The box names the two facing sides it measures and shows today's distance in your units (mm, px or pt — switch under Frame).
3. Type the new distance and press `Enter` (or **Set**). **Sides** picks above / below or left / right if the nearest pair is not the one you want. A negative value overlaps them.

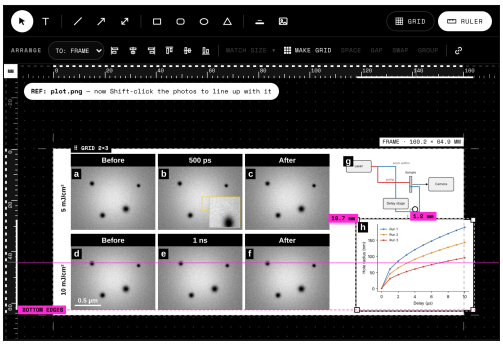
Turn on 🔗 **before** Set to keep the gap: when the reference moves or the grid grows, the other one follows (a pink **LOCKED GAP** label shows it). With more than two selected, each one gets the gap to the reference, on its own side.

Snapping and live distances

While you drag, edges and centres snap to the frame and its margin, to other photos and panels, to guides and — if Snap is on — to the alignment grid. A pink line shows what snapped. Hold `⌘`/`Ctrl` while dragging to move freely.

Pink labels show the distance to the nearest thing on each side (up to 30 mm away) in your units, so you can match gaps by eye.

`Shift` while dragging keeps the movement horizontal or vertical. Arrow keys nudge by 0.1 mm (or 1 px); with `Shift` by 1 mm, or one alignment-grid step when the grid is on.



While dragging, pink labels give the gap to the nearest neighbours.

07

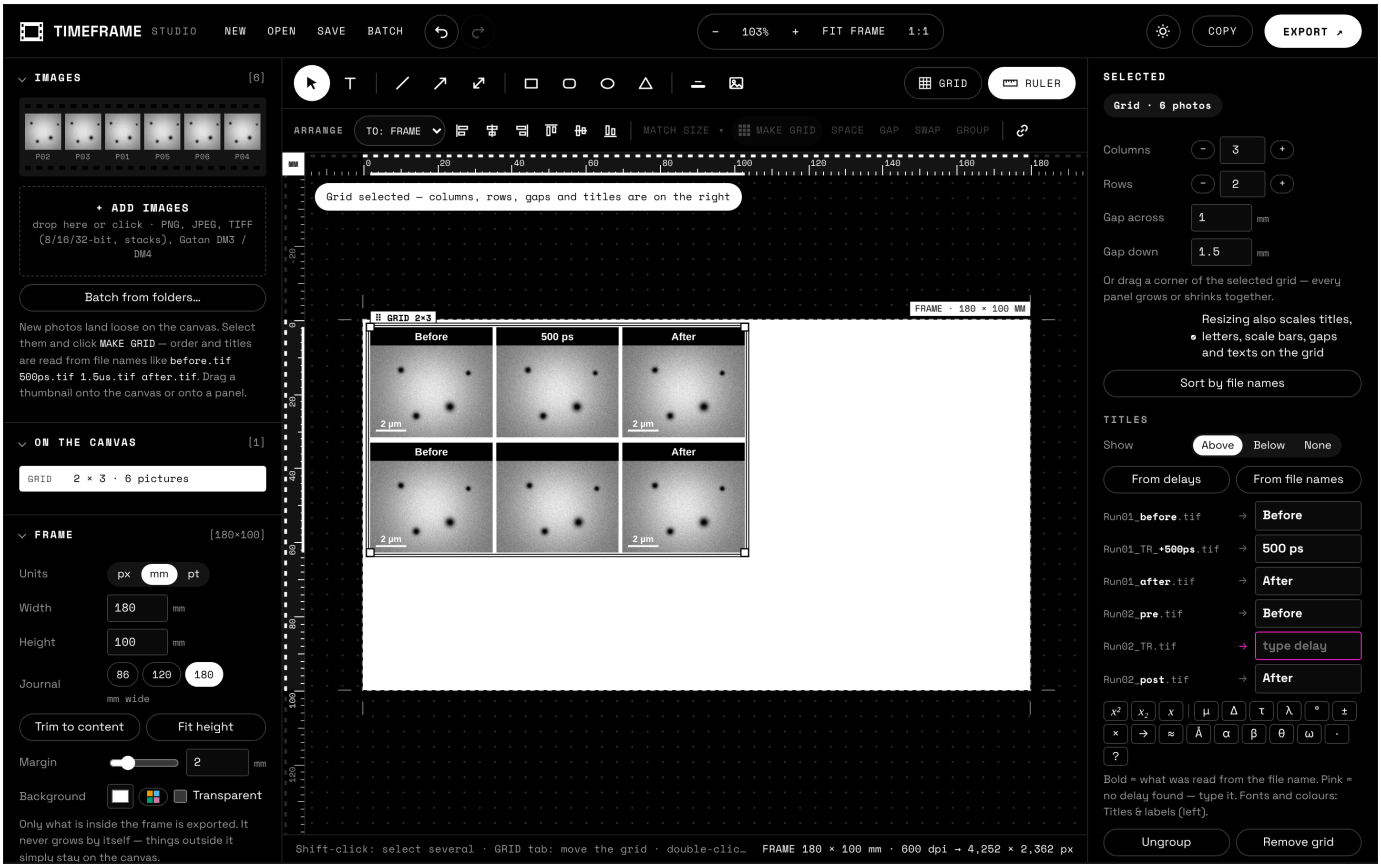
Time-series grids

A grid keeps Before, time-resolved and After photos in tidy rows and columns with titles. It is optional — use it when it helps, and leave other pictures loose.

Making a grid

1. Select the photos — click the one whose **size** you want first (it becomes the reference), then Shift-click the rest, or drag a box around them.
2. Click **`MAKE GRID`**. The grid appears straight away and is selected; its settings are on the right.

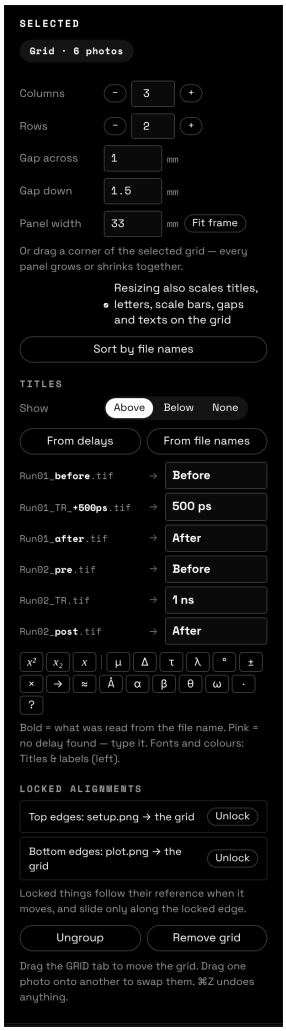
- **Order**: Before → shortest delay → longest → After. If the names show several runs (`Run01_…` , `Run02_…`), each run gets its own row.
- **Titles** come from the delays. A name without one (e.g. `Run02_TR.tif`) gets an empty title, shown in **pink** — type the delay.
- **Size**: every panel takes the width of the reference photo. If that would be wider than the frame, the grid fits the frame instead.
- **One grid per figure**: selecting more loose photos and clicking MAKE GRID again adds them to the existing grid.



Right after MAKE GRID: two runs, one per row, titles filled in; the empty pink title waits for its delay.

Grid settings

- **Columns / Rows** — the − and + buttons; the other number follows so every photo has a place.
- **Gap across / Gap down** — the space between panels, in your units (0 is allowed).
- **Panel width** — type it; or click **Fit frame** to share the frame's width.
- **Sort by file names** — puts them back in Before → delays → After order.
- **Titles** — Above, Below or None; **From delays** or **From file names** refill them all; each title can be typed in the table. Bold shows what was read from the name.
- **Ungroup** — every panel becomes a loose photo again (titles and scale bars become normal objects, contrast stays the same). **Remove grid** takes it off the canvas; its photos stay in the library.



With the grid selected, its settings are on the right.

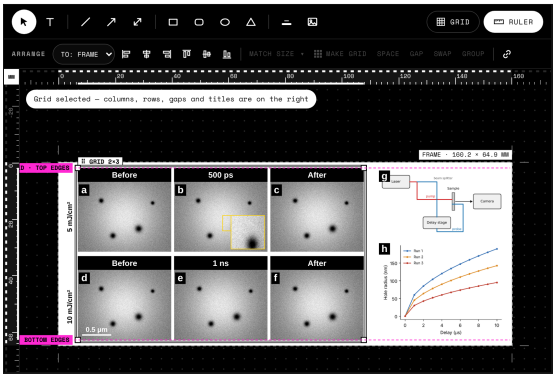
Moving and resizing

- **Move**: drag the ⠿ `GRID` tab above its top-left corner. While the grid is selected you can also drag it by any panel, or nudge it with the arrow keys. Everything that belongs to the grid comes along: texts and arrows on its panels, labels copied with Repeat (also those beside it) and free scale bars.
- **Resize**: with the grid selected, drag one of its **four corner handles** — all panels grow or shrink together and the opposite corner stays put. Titles, letters, row labels, scale bars, gaps and the texts on the grid scale with it, like zooming a picture. Untick **Resizing also scales…** (grid settings) to keep text sizes fixed.
- **Line it up** with other things like any object: e.g. click the grid's tab first (REF), Shift-click a schematic, **align top edges**.

Photos inside a grid

- **Click** a panel to edit its title, crop, contrast and pixel size (chapter 08). The hint at the top says it is part of a grid, with **Select grid** and **Ungroup** buttons.
- **Drag** a photo onto another panel to swap them.
- **Double-click** a title to type it — Tab jumps to the next panel. Double-click an image to crop it.
- **Pop out of grid** (in the panel's settings) turns one panel into a loose photo.

More, in the left panel (TIME-SERIES GRID): Native 1:1 panels, panel shape (square, 4:3, 16:9…), image border, highlight width, Smooth or Pixel-exact enlarging.



A selected grid: the GRID tab is highlighted and the white corner handles resize every panel at once.

08

Panel settings

Text and order

- **Label 1** is the title; **Label 2** a second line (e.g. “single shot”). Under the boxes: **x^2**, **x_2** and the italic ***x*** for formulas (chapter 12), and buttons that insert μ, Δ, τ, λ, °, ±, ×, →, ≈, Å, α, β, θ, ω, ·.
- **Highlight** — a coloured frame and title colour for this panel.
- **Letter** — type your own letter, or tick *none* (when panel letters are on).
- **◀ Move / Move ▶**, **Replace…**, **Remove**, and **Pop out of grid**.
- **Turn & flip** NEW — ↺ 90° and ↻ 90° turn this photo, ⇋ and ⇵ mirror it. **Lossless**: pixels are re-ordered, never resampled. A note then reminds you to mention it in the legend if orientation matters; **Undo turns** puts it back.

Crop & zoom

- **Zoom** slider; **Drag to position…** (or double-click the image), then drag and scroll; **Reset**.
- **Same crop for every panel** (on by default) keeps the same field of view in all panels.
- **+ Zoom inset** — chapter 13.

Brightness & contrast

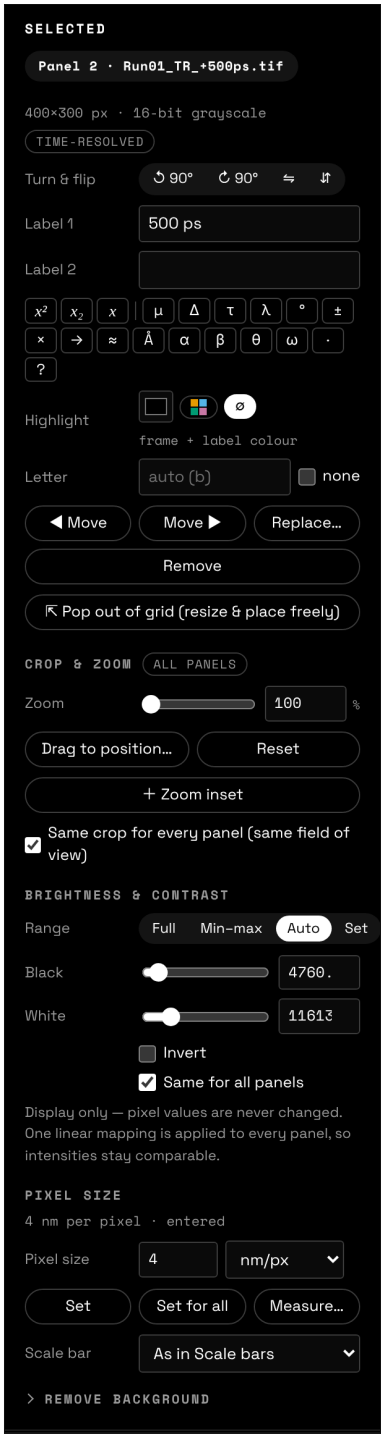
- **Full** — the whole bit range, exactly as recorded. **Min–max** — darkest pixel black, brightest white. **Auto** — clips 0.2 % at each end. **Set** — your own black and white points.
- **Same for all panels** applies one linear mapping to every panel, so intensities stay comparable.
- Display only — the pixel values in your files are never changed.

Pixel size

Read from the TIFF when available. Otherwise type it and click **Set** or **Set for all**, or click **Measure…**, drag across a feature of known length and type that length.

Scale bar and background

Scale bar: as set in Scale bars (left), always here, or never here. **Remove background**: for schematics and renders on a plain backdrop — not for data panels.



Click a panel of the grid to see its settings.

09

Titles, row labels and panel letters

These sections of the left panel set the look for the whole figure, so every panel matches.

Titles & labels

- **Position**: a bar above or below the image, inside one of six spots on the image, or hidden.
- **Show on**: rows that have text (empty bars disappear, so rows without titles sit flush), every row, top row only or bottom row only.
- **Font**, size, bold, italic; **text and bar colours**; bar on or off.
- **Same text → All panels** types one text into every panel. **⇅ Swap** exchanges the positions of label 1 and 2.

Row labels

Vertical labels down the left side of the grid — one sample, fluence or temperature per row. Type them in the list or double-click them on the figure (Tab = next row). They read bottom-to-top like a y-axis label. Height can follow the image or include the title bars. (You can also use a rotated text and **Repeat → This column**, chapter 11.)

Panel letters

- **Style**: a b c · A B C · (a) (b) (c) · a) b) c) · (A) (B) (C) · A. B. C.
- **Position**: inside any corner, outside above the panel, or outside to its left.
- **One per** panel, row or whole grid; order row by row or column by column; skip empty panels.
- **Font**, colour, and an optional **box** behind the letter (useful on light pictures); **Nudge** to fine-tune.
- Loose photos and insets continue the sequence (g, h…). Untick **Letter → show** on a picture to leave it out, or type your own letter.

LAB STYLE

Fonts, colours, letters and scale-bar style are remembered on your computer, so your next figure starts the same way. Share them with your group as a style file (chapter 15).

10

Scale bars

On grid panels (left panel → SCALE BARS)

- **Show on**: every panel, the first, the bottom-left, the last, or none (each panel can override this).
- **Calibrated**: type the length and unit (pm, Å, nm, µm, mm) — the bar is computed from each image's pixel size and the current zoom. **Suggest** picks a round length about ⅕ of the panel width. Panels without a pixel size show no bar, and the checks tell you which.
- **Manual**: you set the bar length and the label yourself. Double-check it — the checks remind you.
- **Corner**, bar colour and thickness, distance from the edge, text above or below, font.

A free scale bar anywhere

Pick the scale-bar tool (B) and drag. In its settings, **Size bar from pixel size** sets its length from a calibrated panel you choose.

ACCURACY

Calibrated bars stay correct when you crop, zoom or resize panels. Manual bars do not — after changing sizes, check them again.

11

Text, arrows, shapes — and Repeat



The toolbar: select, text, line, arrow, double arrow, rectangle, rounded rectangle, ellipse, triangle, scale bar, insert image — then the GRID and RULER switches.

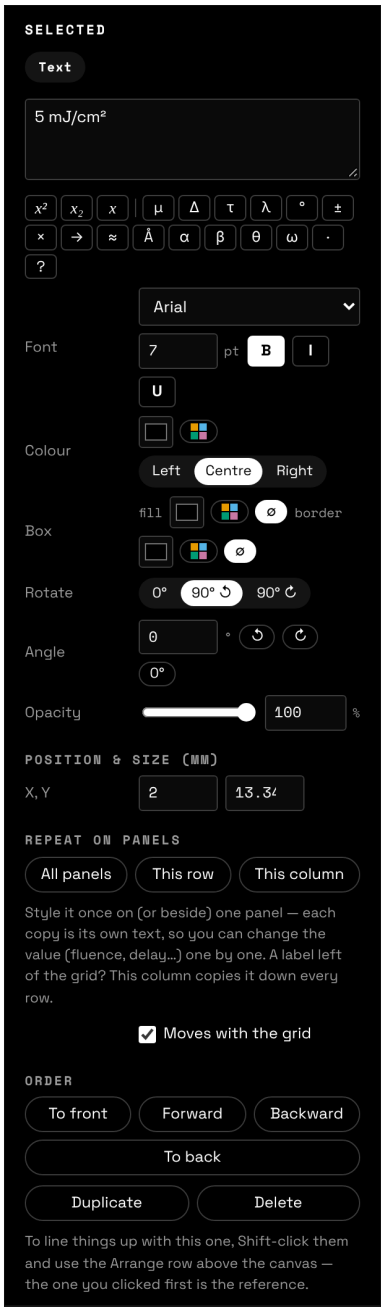
- **Text** (T): click and type; or double-click empty canvas. Font, size, bold / italic / underline, colour, alignment, a box fill and border, 0° or 90° either way, or any angle. Superscripts, subscripts and Greek letters: chapter 12.
- **Line** (L), **arrow** (A), **double arrow**, **rectangle** (R), rounded rectangle, **ellipse** (O), triangle. Shift = square, circle or 45°. Outline, fill, dashes, head size, opacity.
- **Connect to**: a rectangle or ellipse can draw connector lines to a picture or an inset.
- **Order**: To front, Forward, Backward, To back. **Duplicate** (⌘D, or Alt-drag), copy / paste (⌘C / ⌘V), **Delete**.

Repeat on panels

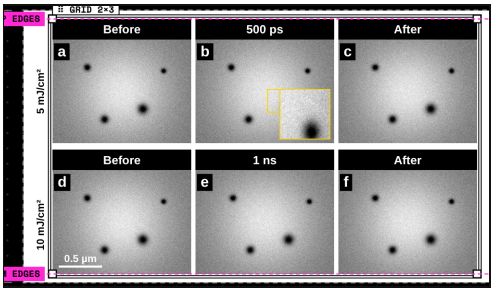
Style a text, arrow or shape once, on one panel or right beside it, then:

- **All panels** — the same spot on every photo of the grid.
- **This row** — only along its row. **This column** — only down its column.
- A label **beside** the grid (e.g. a fluence left of a row) is placed relative to the nearest panel, so **This column** copies it next to every row.

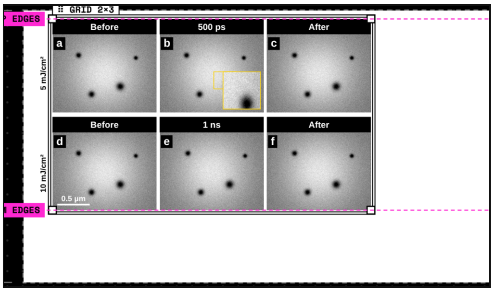
Each copy is its own object: double-click one to change its value (5 → 10 mJ/cm²). The copies **belong to the grid**: they move with it and scale when it is resized. Any text or shape can be attached or detached with **Moves with the grid** in its settings.



A selected text: style, rotation and Repeat on panels.



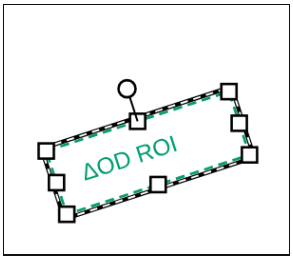
Repeat copies beside the rows, an inset and a scale bar…



…grid made 25 % smaller: all of it scales with the grid.

Turning things NEW

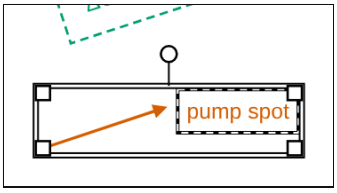
- Select a text, shape or picture and drag the **round handle** above it. `Shift` turns in 15° steps; near 0°, 90°, 180° it snaps straight.
- Or type an **angle** in its settings, with ↺ / ↻ for 90° steps and **0°** to straighten.
- Turned things are lined up and spaced by their outer box, like PowerPoint. Their corner handles still resize them.
- For a photo, prefer **Turn & flip** (lossless, chapter 08). Any other angle resamples the pixels.



Drag the round handle to turn.

Groups NEW

- Select two or more things (Shift-click or drag a box) and click **GROUP** in the Arrange row, or press ⌘G.
- A click on any member selects the whole group. It moves, aligns and spaces as one. A **corner** resizes everything in it, text and line widths included; the **round handle** turns it.
- Double-click a text in a group to edit it. **UNGROUP** or ⇧⌘G makes the items separate again. Groups stay groups in PowerPoint export.



A group: one box, corner handles, a round handle to turn it.

Colours NEW

Every colour setting has a small four-colour button next to it with ready-made colours:

- **Basic** — black, white, greys and a few strong colours.
- **Colour-blind safe** — the Okabe–Ito set, readable by most people with colour-vision deficiency.
- **LUT colours** — the ImageJ channel colours (red, green, blue, cyan, magenta, yellow, grey), and steps of the **Fire**, **Viridis** and **Magma** lookup tables, to match colour-coded images.
- **Recent** — the colours you picked last.

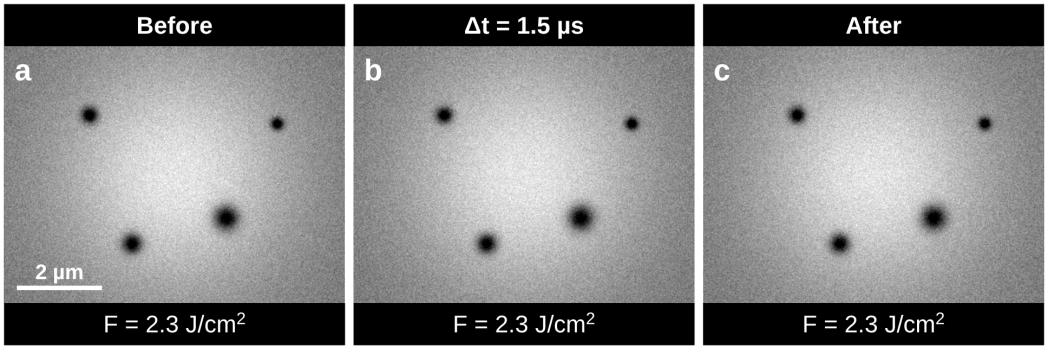


The palette button beside a colour.

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Formulas and symbols

Any text in the figure can hold a formula: titles, second labels, row labels, free text, scale-bar labels and letters. You type a small part of LaTeX; TimeFrame Studio draws it in the text's own font, size, weight and colour, so formulas match the rest of the figure, as journals ask.



$Ge_2Sb_2Te_5$ film · T = 300 K · $\tau_{1/2}$ = 40 ps · 5 × 10^{-3} counts per pulse

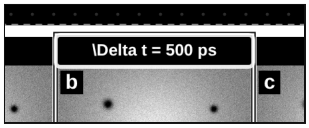
Typed: titles `Before`, `\Delta t = 1.5 \mu s`, `After`; second labels `F = 2.3 J/cm^2` (one typed, then **Same text → All panels**); the line below `Ge_2Sb_2Te_5 film · $T$ = 300 K · \tau_{1/2} = 40 ps · 5 \times 10^{-3} counts per pulse`. Exported PNG, 400 dpi.

What you type → what you get

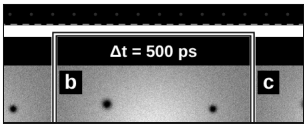
YOU TYPE	YOU GET	NOTES
`x^2` · `cm^-1` · `10^{-3}`	$x^2 \cdot cm^{-1} \cdot 10^{-3}$	Superscript. One character or one whole number; use `{ }` for more.
`H_2O` · `Ge_2Sb_2Te_5`	$H_2O \cdot Ge_2Sb_2Te_5$	Subscript.
`\tau_{1/2}` · `SO_4^{2-}`	$\tau_{1/2} \cdot SO_4^{2-}$	A subscript and a superscript on the same letter are stacked.
`\Delta t = 1.5 \mu s`	Δt = 1.5 µs	Greek letters by name.
`$t$ = 0` · `$\Delta t$`	$t = 0 \cdot \Delta t$	`$ $` makes letters italic, as in LaTeX math.
`20 \pm 2 \degree C`	20 ± 2 °C	
`5 \times 10^{-3}`	5×10^{-3}	Exponents get a real minus sign (−).
`\lambda = 800 nm`	λ = 800 nm	
`\frac{1}{2}` · `\sqrt{2}`	1/2 · √2	Written on one line; no stacked fractions.
`Run\_01` · `\$5` · `\\`	Run_01 · $5 · \	A backslash in front gives the plain character.

On the canvas

While you type a title or a text on the canvas you see the code; press `Enter` or click away and you see the formula. Double-click to change it again.



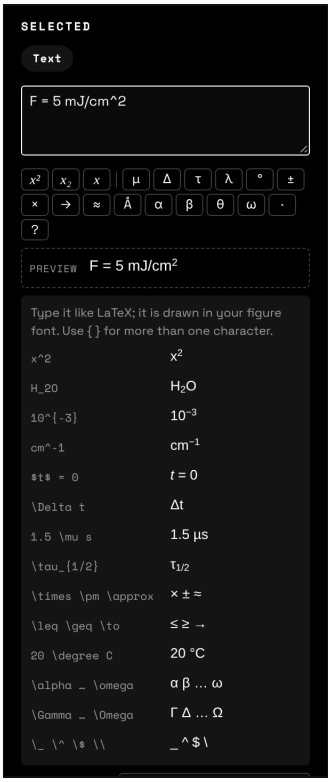
Typing: the code.



Done: the formula.

The buttons and the preview

- **x²** and **x₂** wrap the selected text in `^{ }` or `_{ }`. With nothing selected they insert the braces and put the cursor inside.
- The italic ***x*** wraps the selection in `$ $`.
- The symbol buttons (μ Δ τ λ ° ± × → ≈ Å α β θ ω ·) insert the character itself. Both ways give the same result.
- **PREVIEW** appears under the boxes as soon as a text holds formula code, and shows what will be printed.
- **?** opens a short help card with examples.



A selected text with a formula: the formula buttons, the preview, and the help card from **?**.

Good to know

- **Plain text is unchanged.** Text without `\ ^ _ $` is drawn exactly as before.
- **Spaces after Greek letters.** After `\mu` and `\degree` the space before a unit is dropped (`\mu s` → μs, `\degree C` → °C). After other Greek letters it is dropped before a one- or two-letter symbol (`\Delta t` → Δt, `\Delta OD` → ΔOD) and kept before a word (`\alpha phase` → α phase). Type `\ ` (backslash, space) to force a space.
- **Inside** `$ $` letters and small Greek letters are italic; numbers, units and capital Greek stay upright; `-` becomes a real minus. Spaces stay as you type them.
- **Typos**: an unknown command such as `\taue` is printed as typed. The preview names it, and so do the checks before export.
- **File-name titles and older projects**: titles read from file names are stored with `\_` so they print exactly as written. Projects saved with version 19 or earlier are converted the same way when opened, so they look the same.
- **Export**: sharp at any dpi; in SVG they stay editable text.

All commands

GROUP	COMMANDS
Greek	`\alpha` α `\beta` β `\gamma` γ `\delta` δ `\epsilon` ε `\zeta` ζ `\eta` η `\theta` θ `\iota` ι `\kappa` κ `\lambda` λ `\mu` μ `\nu` ν `\xi` ξ `\pi` π `\rho` ρ `\sigma` σ `\tau` τ `\upsilon` υ `\phi` φ `\chi` χ `\psi` ψ `\omega` ω
Capitals	`\Gamma` Γ `\Delta` Δ `\Theta` Θ `\Lambda` Λ `\Xi` Ξ `\Pi` Π `\Sigma` Σ `\Phi` Φ `\Psi` Ψ `\Omega` Ω
Operators	`\times` × `\cdot` · `\pm` ± `\mp` ∓ `\div` ÷ `\approx` ≈ `\sim` ∼ `\propto` ∝ `\equiv` ≡ `\neq` ≠ `\leq` ≤ `\geq` ≥ `\ll` ≪ `\gg` ≫ `\minus` −
Arrows	`\to` → `\leftarrow` ← `\leftrightarrow` ↔ `\Rightarrow` ⇒ `\uparrow` ↑ `\downarrow` ↓ `\rightleftharpoons` ⇌
Other	`\degree` ° `\AA` Å `\hbar` ħ `\ell` ℓ `\infty` ∞ `\partial` ∂ `\nabla` ∇ `\sum` ∑ `\int` ∫ `\ldots` … `\prime` ′ `\sqrt{x}` √x `\frac{a}{b}` a/b
Spacing	`\,` thin space · `\ ` normal space · `\quad` wide space
Style	`\text{…}` upright · `\textit{…}` italic · `\textbf{…}` bold
Plain	`\_` _ `\^` ^ `\$` $ `\\` \ `\{` { `\}` } `\%` %

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Zoom insets and free pictures

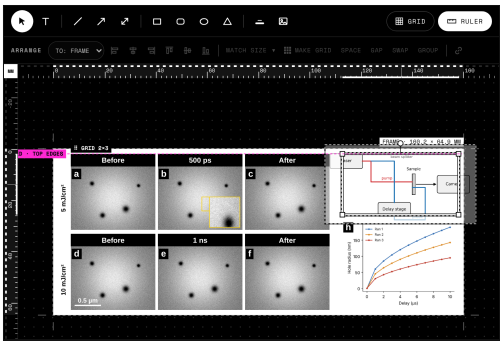
Zoom insets

1. Click a panel, then **+ Zoom inset**.
2. Drag a box over the area to magnify. The inset appears in the panel's corner.
3. Drag the inset anywhere — on the panel or beside it. Drag the dashed box on the panel to move the magnified area; its corner handle resizes it.

Settings: frame colour and width, line style, the box on the panel, connector lines (shown once the inset sits beside the box), and Smooth or Pixel-exact enlarging.

Free pictures: schematics, renders, plots

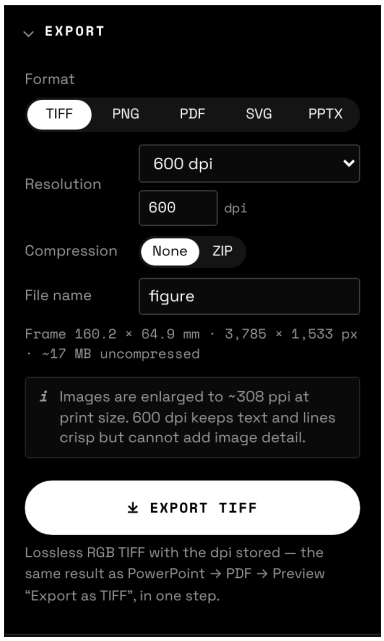
- Drop them like photos, or use the **insert image** tool.
- **Resize** from a corner (keeps proportions; `Shift` to stretch). **Turn & flip** in 90° steps is lossless; **Rotate** takes any angle but resamples the pixels.
- **Double-click** a picture (or click **Choose what shows…**) to slide it inside its box and scroll to zoom — useful after Match size. **Show all** undoes the crop.
- **Crop edges** sliders, a **border**, its own **brightness & contrast** and **background removal**.
- Line them up with the Arrange row, and lock the alignment so they follow the grid when you change its gaps or size.



Choose what shows: drag the picture inside its box, scroll to zoom. The faded part is cropped away.

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Exporting



Export, with the checks above the button.

FORMAT	BEST FOR
TIFF	Journal submission. Lossless RGB (RGBA if transparent) with the dpi stored. Compression **None** opens everywhere; **ZIP** gives identical pixels in a smaller file.
PNG	Lossless with the dpi stored; smaller files; web and slides.
PDF	One page at the exact print size, the figure embedded losslessly at your dpi.
SVG	Illustrator / Inkscape: editable text and shapes, photos embedded at their own resolution.
PPTX NEW	PowerPoint (also Keynote, Google Slides, LibreOffice): one slide at the figure's exact size. Text stays text, arrows and shapes stay shapes, pictures are embedded losslessly at their own resolution.

GOOD DEFAULT FOR A PAPER

Frame at the journal's column width · TIFF · 600 dpi · Compression None (or ZIP) · no warnings left in the checks.
Camera pixels untouched: put the photos in a grid and set **Panel size → Native 1:1** (left, Time-series grid) — one camera pixel = one exported pixel. Add **Pixel-exact** enlarging and **Full** contrast to show the data as recorded. Loose pictures are always scaled to their box.

- **Resolution**: 300, 600 or 1200 dpi, or any value. 600 dpi suits figures that mix images and text; many journals ask ~300 dpi for photos and 1000–1200 for line art — check yours.
- **Only the frame is exported.** Anything outside it is left out.
- **COPY** puts a PNG on the clipboard for slides — limited to 300 dpi. Use EXPORT for papers. ⌘E exports.

PowerPoint export NEW

- The **grid is one group**, with a sub-group per panel (title bars, picture, letter, scale bar) and per row label. Your own groups stay groups; turned things keep their angle.
- Formulas become text with real superscripts and subscripts, which PowerPoint draws at its own size — slightly smaller than in the figure. A subscript and superscript on the same letter (SO_4^{2-}) sit side by side.
- Fonts are not embedded: the computer opening it needs them (Arial is safe). Dashes use PowerPoint's own dash patterns, and enlarged pictures are always smoothed (there is no Pixel-exact there).
- For the journal, export TIFF from TimeFrame Studio; the slide is for talks and for people who want to change things in PowerPoint. Changes made there do not come back into the project.

The checks

Before you export, the list above the button tells you about: panels without a pixel size, manual scale bars, text smaller than 5 pt, images enlarged beyond their real resolution, contrast set per panel, background removal on data, items crossing the frame edge or lying outside it.

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Saving, sharing and your lab style

Projects

- **SAVE** (⌘S) downloads a project file, e.g. `figure1.tfs.json`. It contains everything — layout, text, settings and your **original image files, unchanged** — so it can be reopened anywhere.
- **OPEN** (⌘O) loads a project. It replaces what is on screen, so save first.
- **NEW** starts an empty figure and keeps your style and frame size.
- Projects from older versions open too. Pictures that used to be docked beside the grid become loose pictures in the same place.

Sending a figure to your PI or co-authors

1. Click **SAVE** to get the `.tfs.json` project.
2. Send it **together with the TimeFrame Studio HTML file** you used (the same version), and the exported TIFF for reference.
3. They double-click the HTML (it opens in their browser), click **OPEN**, choose the project, edit, click **SAVE** and send the new file back — ideally renamed (e.g. `figure1_edited.tfs.json`).

> FONTS
> If you use a font they do not have (Calibri on a Mac without Office, for example), their screen shows a substitute. Arial and Helvetica look the same everywhere.

For Illustrator or Inkscape users, export **SVG** instead: text and shapes stay editable, but the grid becomes separate pieces. For someone who works in **PowerPoint**, export **PPTX**: they can change words, colours and positions there. Either way, keep the `.tfs.json` as your master copy.

Lab style

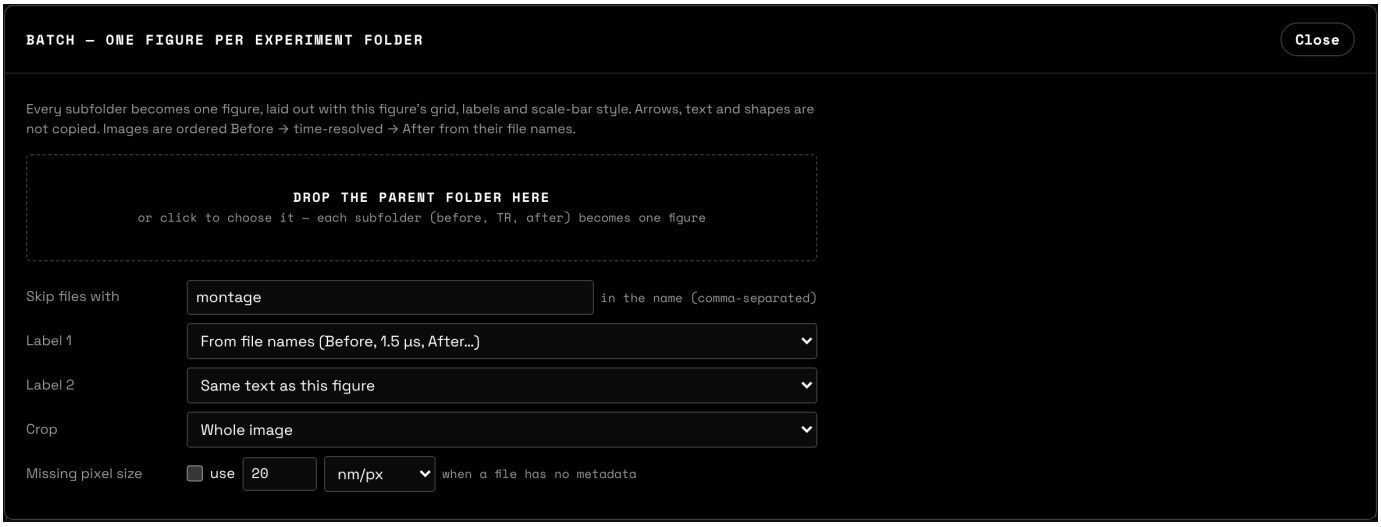
Your label, letter and scale-bar style is remembered on this computer. In **LAB STYLE** (left) you can **Save style file** to share it with your group, **Load style file** to use someone else's, or **Reset** to the defaults.

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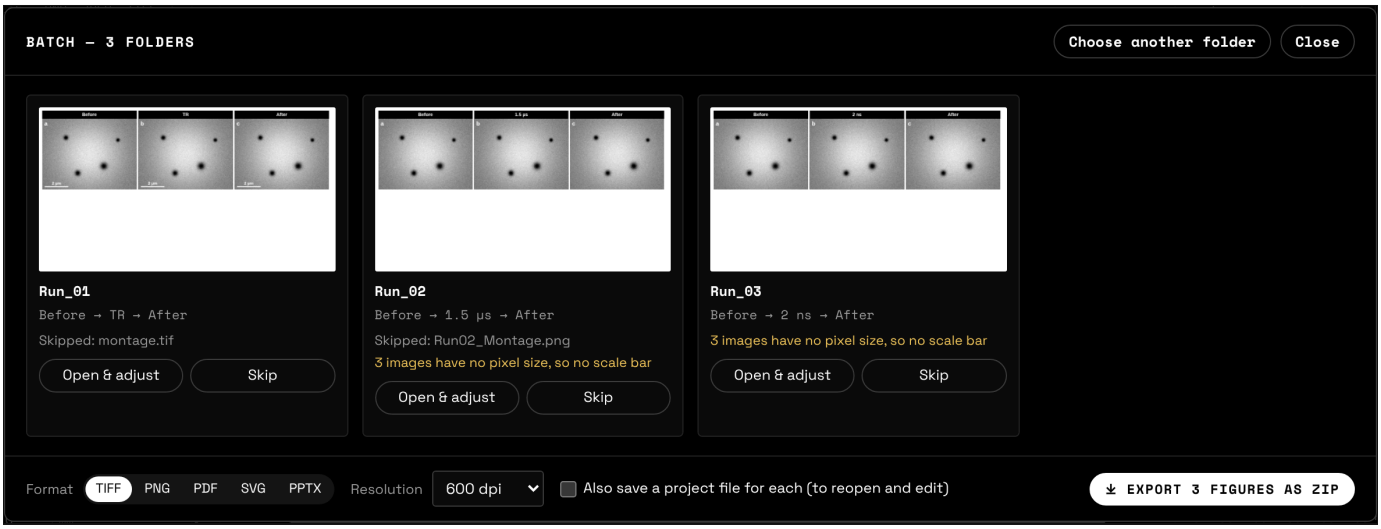
Batch: one figure per experiment

Make one figure the way you like it, then build the same figure for every experiment folder in one go.

1. Finish a template figure — grid, titles, letters and scale bars as you want them.
2. Click **BATCH** and drop the **parent folder** (or click to choose it). Each subfolder becomes one figure.
3. Choose: files to **skip** (e.g. `montage`), where **Label 1 / 2** come from (file names, the template's text, or empty), the **crop** (whole image, or the template's), and a pixel size for files without metadata.
4. Review the cards: each shows the figure, its order and any warnings. **Open & adjust** one to change it, then **Back to batch**. **Skip** leaves one out.
5. Click **EXPORT … AS ZIP**. Tick *Also save a project file for each* to be able to reopen them.



Batch set-up.



Review: one card per folder, with the order read from the names and any warnings.

Copied from the template: the grid, title and label styles, row labels, panel letters, scale bars and the frame. Not copied: text, arrows, shapes and loose pictures. If the template has no grid, a default one is used.

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Step-by-step recipes

A. A Before · time-resolved · After figure in two minutes

1. FRAME → 180 (or your journal width).
2. Drop all the photos of the series. They arrive selected.
3. Click **`MAKE GRID`**. Type any delay shown in pink on the right.
4. Left: PANEL LETTERS → a b c; SCALE BARS → Calibrated, Suggest; Show on → bottom-left panel.
5. FRAME → **Fit height**. Check the warnings, then **`EXPORT TIFF`**.

B. Two pictures, same size, side by side (BrightSlide style)

1. Click picture A (the one whose size is right), Shift-click picture B.
2. **`MATCH SIZE`** — B becomes exactly A's size (cropped to fill; double-click B to choose what shows). Or ▾ → **Height only** to keep B's proportions.
3. **`Align top edges`**. Then **`SPACE`** → Exact gap 2 mm → ↔.

C. A fluence (or sample) label for every row

1. Text tool: click left of the first row and type "5 mJ/cm²". Rotate 90° ↻, style it.
2. With it selected: **Repeat on panels → This column**. A copy appears beside every row.
3. Double-click each copy to change its value.

D. A schematic that stays lined up with the grid

1. Click the grid's ⠿ **`GRID`** tab (reference), Shift-click the schematic.
2. Click 🔗, then **`Align top edges`**. Now it follows when you change gaps, rows or the grid size.
3. For a picture under it: click the grid tab, Shift-click the picture, 🔗, **`Align bottom edges`**.

E. Make the grid smaller or bigger

1. Click the ⠿ **`GRID`** tab.
2. Drag a white corner handle — or type **Panel width** on the right; **Fit frame** to fill the width.

F. A figure exactly 86 mm wide (one column)

1. FRAME → 86. Select the grid → Panel width → Fit frame.
2. Line up the rest; FRAME → Fit height. Keep text ≥ 6 pt (check the warnings).

G. Exact camera pixels for a methods figure

1. Make the grid. Left → TIME-SERIES GRID → Panel size → **Native 1:1**; Enlarging → Pixel-exact.
2. FRAME → **Trim to content**. Export TIFF — every camera pixel is one exported pixel.

H. A 2 mm gap under the grid that stays 2 mm

1. Click the ⠿ **GRID** tab, Shift-click the picture below it.
2. Click 🔗, then **GAP**. Units → mm (Frame, left) if needed. Type 2, press `Enter`.
3. Resize or move the grid: the picture stays 2 mm below it.

I. An editable PowerPoint for your PI or a talk

1. Export → Format **PPTX** → **EXPORT PPTX**.
2. Open it in PowerPoint: click into any title or label to change it; the grid moves as one group (ungroup it there to move single panels).

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Keyboard and mouse

On Windows and Linux use `Ctrl` where this says `⌘`.

KEY	ACTION
`V`	Select
`T`	Text
`L` / `A`	Line / arrow
`R` / `O`	Rectangle / ellipse
`B`	Scale bar
`G`	Alignment grid on / off
`⇧R`	Rulers on / off
`⌘Z` / `⇧⌘Z`	Undo / redo
`⌘A`	Select all
`⌘C` / `⌘V` / `⌘D`	Copy / paste / duplicate
`⌘G` / `⇧⌘G`	Group / ungroup
`Delete`	Delete the selection (a grid: remove it)
Arrow keys	Nudge (with `Shift`: bigger step)
`Esc`	Leave crop / tool / pop-up; clear the selection
`⌘S` / `⌘O` / `⌘E`	Save / open / export

MOUSE	ACTION
Click	Select (the first = reference)
`Shift` / `⌘`-click	Add or remove
Drag on empty canvas	Box-select
Drag	Move (snaps; `⌘` = free; `Shift` = straight)
`Alt`-drag	Duplicate
Drag panel → panel	Swap photos
Drag GRID tab	Move the grid
Drag corner	Resize (a grid or group: everything on it)
Drag round handle	Turn (`Shift` = 15° steps)
Double-click picture	Choose what shows / crop
Double-click title	Type it (`Tab` = next)
Double-click empty	New text
`⌘` + scroll	Zoom around the pointer
Drag from a ruler	New guide

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Good practice for scientific figures

- **Same contrast for comparable panels.** Keep “Same for all panels” on for a time series. If you adjust panels separately, say so in the legend — the checks remind you.
- **Calibrated scale bars.** Prefer them to manual ones; they stay right when you crop or resize.
- **Readable text.** At final print size, use at least 5–6 pt (many journals ask for 6–8 pt). Design at the real width, so what you see is what is printed.
- **Don’t invent detail.** 600 dpi keeps lines and text crisp but cannot add information to a 400-pixel image. Native 1:1 shows what the camera recorded.
- **No background removal on data.** Use it for schematics and renders only.
- **Keep the originals.** The project file stores them unchanged — keep it with the exported figure.
- **Italic quantities, upright units.** `$t$ = 1.5 \mu s` → *t* = 1.5 µs; `$\Delta t$` → Δ*t*.
- **Be consistent.** Use the lab style and Batch so all figures of a paper share fonts, letters and spacing.

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Troubleshooting

WHAT YOU SEE	WHY, AND WHAT TO DO
The Arrange buttons are grey	Nothing (or not enough) is selected. Click the reference, Shift-click the others.
Align or Match size does nothing to a photo	It is a panel of the grid — panels move together. Select the grid, or **Ungroup** it to arrange photos one by one.
The grid came out too big or small	It takes the width of the photo you clicked first. Drag a corner of the selected grid, or type Panel width.
A picture won’t move up or down	It has a locked alignment. Unlock it in the right panel.
Something is cut off in the export	It crosses the frame edge. Move it, enlarge the frame, or use Trim to content.
No scale bar on some panels	Those images have no pixel size. Set it (panel settings → Pixel size) or use a manual bar.
Repeat says “Put it on a panel…”	The text is too far from the grid. Move it onto, or right next to, a panel first.
Part of a text turned small and low or high	`_` and `^` make sub- and superscripts; type `\_` or `\^` (chapter 12).
Text looks different on another computer	That computer lacks the font. Use Arial or Helvetica.
Export refuses (“over 600 megapixels”)	Lower the dpi or the frame size.
COPY doesn’t work	Some browsers can’t copy images. Use Export; Chrome, Edge and Safari can copy.

LIMITS

One grid per figure. Text in the figure uses the fonts installed on the computer. The exported figure is 8-bit RGB — your 16-bit originals live on in the project file.